ELEC3104: Digital Signal Processing

Topic 2: Discrete-Time Systems
Tutorial questions

Prof. Eliathamby Ambikairajah, School of EE&T

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Topic 2: Pass- Level Tutorial Questions

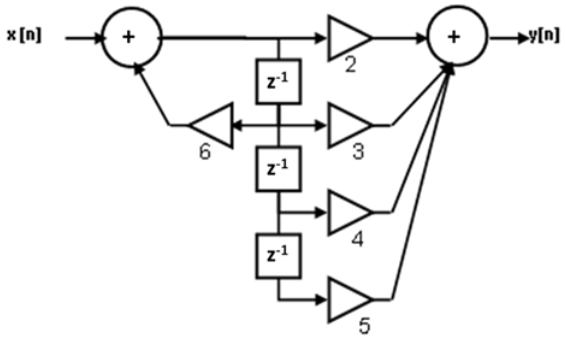
1. What is the equivalent transfer function of the block diagram below? (Tick the correct answer)

(a) $H(z) = \frac{z(z^{-1}+3)(z^{-1}+4)(z^{-1}+5)}{6-z^{-1}}$

(b) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1-6z^{-1}}$

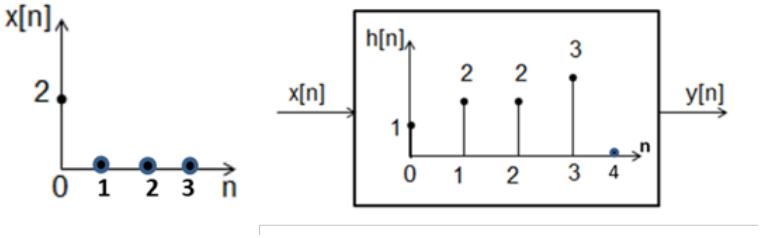
(c) $H(z) = \frac{1+6z^{-1}}{2+3z^{-1}+4z^{-2}+5z^{-3}}$

(d) $H(z) = \frac{2+3z^{-1}+4z^{-2}+5z^{-3}}{1+6z^{-1}}$



2. The impulse response ,$h[n]$, of a discrete-time system is given below. The output signal, $y[n]$, for the input signal, $x[n]$ is given by: (Tick the correct answer)

(a) {2 0 0 0 0}
(b) {1 2 2 0 0}
(c) {3 4 5 2 0}
(d) {2 4 4 6 0}
(e) None of the above



3. The impulse response of the system with the transfer function $H(z) = \frac{\alpha+\alpha z^{-1}}{1-\alpha z^{-1}}$ is given by: (Tick the correct answer)

(a) $\alpha^{n-1} + \alpha$ n < 0

(b) α^{n} u[n] + α^{n-1} u[n-1]

(c) α^{n+1} u[n] + α^{n} u[n-1]

(d) None of the above

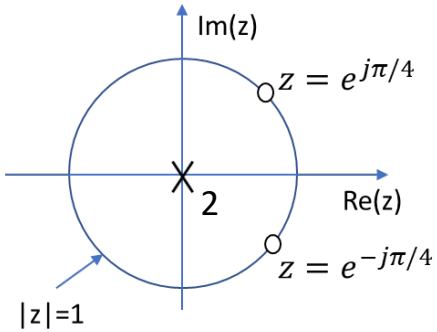
4. The pole-zero pattern of a system has zeros on the unit circle as show below. The system function is given by: (Tick the correct answer)

(a) $H(z) = 1 + \frac{1}{\sqrt{2}} z^{-1} + z^{-2}$

(b) $H(z) = z - \frac{1}{\sqrt{2}} + z^{-1}$

(c) $H(z) = z^2 - 1 + z^{-2}$

(d) $H(z) = 1 - \sqrt{2}\, z^{-1} + z^{-2}$



5. The DFT of a real signal is given below. The signal energy is: (Tick the correct answer)

(a) 0
(b) 5
(c) 7
(d) None of the above

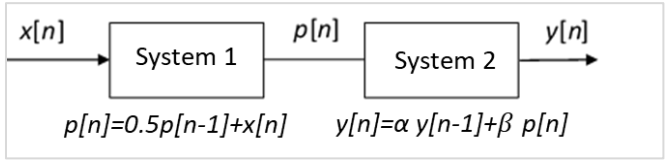
DFT: [1, $\sqrt{2}$, -1, B, -7, -j2, C, $\sqrt{2}$]
k=0

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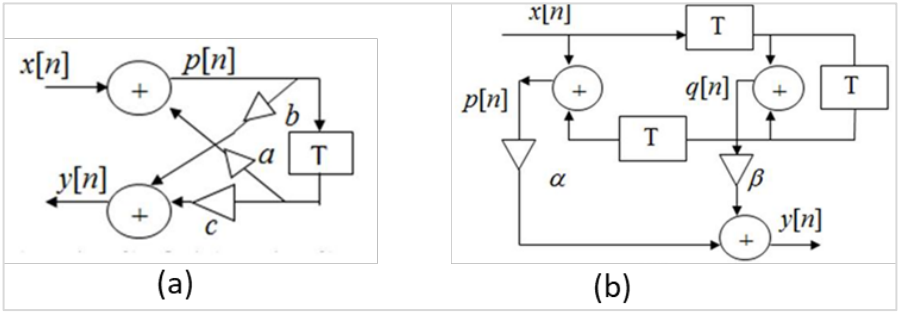
1. Consider the cascade of the following two systems as depicted below. The difference equation relating $x[n]$ and $y[n]$ is given by:

$$y[n] = x[n] + \frac{3}{4}y[n-1] - \frac{1}{8}y[n-2]$$

Determine α and β.



2. For the block diagram realisations given below, develop the relation between y[n] and x[n] in each case.



3. Draw a system implementation for each of the following equations:

a) $2y[n] + y[n-1] - 4y[n-3] = x[n] + 3x[n-5]$

b) $y[n] = x[n] - x[n-N]$

c) $y[n] = a_0x[n] + a_1x[n-1] + a_2x[n-2] - b_1y[n-1] - b_2y[n-2]$

4. A difference equation of a particular filter is given by

$$y[n] = 0.1x[n] - 0.1x[n-2] + 0.8x[n-3] + 0.1x[n-4] + 0.6x[n-6]$$

Find the impulse response of the above filter.

5. Consider the stable causal system described by the difference equation

$$y[n] = ay[n-1] + bx[n] \qquad |a|<1$$

Determine '*b*' in terms of '*a*' so that $\sum_{n=-\infty}^{\infty} h[n] = 1$.

6. Let $x[n]$ = {1 4 0 2} and $h[n]$ = {1 2 1}. Find their convolution (Both sequences start at $n = 0$).

7. Find the impulse response $h[n]$ for the following causal system:

$$H(z) = \frac{1 - \frac{1}{2}z^{-1}}{\left[1 - \frac{1}{4}z^{-1}\right]\left[1 + \frac{1}{4}z^{-1}\right]}$$

8. Consider the following system:

$$H(z) = \frac{1 - 2z^{-1} + 2z^{-2} - z^{-3}}{(1 - z^{-1})(1 - 0.5z^{-1})(1 - 0.2z^{-1})}$$

a) Sketch the pole-zero pattern. Is this system stable?
b) Determine the impulse response of this causal system.

9. Determine and sketch the approximate magnitude response for the following filter: $H(z) = \frac{1-a}{1-az^{-1}}$ 0<a<1

10. A speech signal is sampled with a sampling period of 125μs. A frame of 256 samples is selected and a 256 point DFT is computed. What is the spacing between the DFT values in Hz.

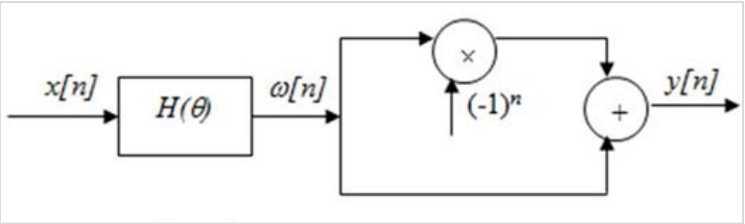
Topic 2: Pass – Level Tutorial Questions

11. Let x[n]={1, 2, 1, 0}. With N=4, compute the 4-point DFT.

12. Find the 8-point DFT of $x[n]$ ={1, 1, 0, 0, 0, 0, 0, 0}, using conjugate symmetry property of $X[k]$.

13. For the system in the figure below, sketch the output *y[n]* when the input *x[n]* is $\delta[n]$ and $H(\theta)$ is an ideal lowpass filter as follows:

$$H(\theta) = \begin{cases} 1 & 0 \le |\theta| \le \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < |\theta| \le \pi \end{cases}$$



14. A 3- second signal is sampled at f_s = 100 Hz. The maximum spectral spacing is to be $\Delta f = 0.25\ Hz$. How many samples are needed for the DFT and FFT ?

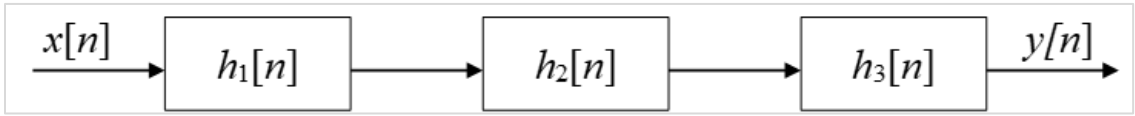
15. Classify the following system in terms of its linearity, time invariance, memory and causality: $y[n] = |x[-n]|$

16. Consider a system with an input x[n] and an output y[n] which satisfies the following difference equation:

$$y[n] = n\,y[n-1] + x[n]$$

If $x[n] = \delta[n]$, determine $y[n]$ for all values of n.

17. Three systems with the following impulse responses are connected in cascade: $h_1[n] = \delta(n) - \delta(n-1)$; $h_2[n]$ = g[n]; $h_3[n] = u[n]$; where $u[n]$ is the unit step sequence and $\delta[n]$ is the unit sample sequence (i.e an impulse).



What is the impulse response $h_c(n)$ of the overall system?

18. Find the z-transform, G(z) and its ROC , if $g[n] = \begin{cases} na^n & n \ge 0 \\ 0 & n < 0 \end{cases}$

where *a* is a constant

19. Consider the interconnection of LTI systems as shown in Figure 2 below:

(a) Express the overall impulse response in terms of h_1[n], h_2[n], h_3[n] and h_4[n].

(b) Determine *h[n]* when

h_1[n] = $\{\frac{1}{2}\ \frac{1}{4}\ \frac{1}{2}\}$; h_2[n] = h_3[n] = (n+1) u[n]; h_4[n] =$\delta[n-2]$

(c) Determine the output $y[n]$ of the system in part(b) for the input

$x[n] = \delta[n+2] + 3\delta[n-1] - 4\delta[n-3]$

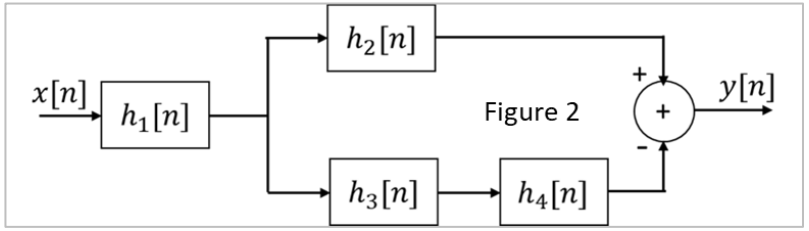


Figure 2

Topic 2: Higher – Level Tutorial Questions

1. If $y[n] = x[n] * h[n]$, show that Y(z) = X(z) . H(z).

2. The frequency response of a band stop filter is given by

$$H_d(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{6} \\ 0 & \frac{\pi}{6} < |\theta| < \frac{\pi}{3} \\ 1 & \frac{\pi}{3} \le |\theta| \le \pi \end{cases}$$

Show that the impulse response $h_d[n]$ of the band stop filter is given by

$$h_d[n] = \delta[n] - \frac{sin\left(\frac{\pi}{3}n\right)}{\pi n} + \frac{sin\left(\frac{\pi}{6}n\right)}{\pi n}$$

3. Show that the convolution of the two infinite duration sequences

$$p[n] = a^n u[n] \quad \text{and} \quad q[n] = b^n u[n]$$

where u[n] is the unit step function and a ≠ b, is given by

$$y[n] = \frac{a^{n+1} - b^{n+1}}{a-b} u[n]$$

4. A system L is known to be linear. The responses of L to inputs $x_1[n]$ and $x_2[n]$ are given by $y_1[n]$ and $y_2[n]$ respectively (see Figure 1).

   (a) Determine whether the system *L* could be time-invariant.

   (b) If the inputto the system is $x[n] = \delta[n]$, what is the system response y[n]?

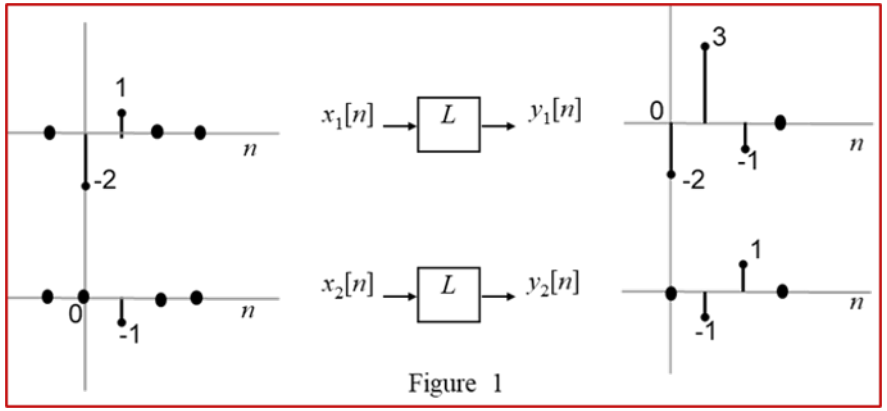


Figure 1

5. Find a difference equation for the system with the impulse response:

$$h[n] = (\tfrac{1}{4})^n \cos(n\tfrac{\pi}{3})\, u[n]$$

6. An Lth order moving average filter is a linear time invariant system that for an input $x[n]$, produces the output $y[n] = \frac{1}{L+1}\sum_{k=0}^{L} x[n-k]$. Show that its frequency response is $H(\theta) = \frac{1}{L+1} e^{-jL\theta/2} \frac{\sin[(L+1)\theta/2]}{\sin(\frac{\theta}{2})}$

Topic 2: Higher – Level Tutorial Questions

7. Consider a bandpass filter with DTFT, $H(\theta)$ as shown in Figure A. Show that $h[n]$ can be expressed as:

$$h[n] = (k_p + k_q)\cos(n\theta_0) - k_p\cos(n\theta_u) - k_q\cos(n\theta_l)$$

$$k_p = \frac{1}{\pi n^2(\theta_u - \theta_0)}; \qquad k_q = \frac{1}{\pi n^2(\theta_0 - \theta_l)}; \qquad \theta = 2\pi\frac{f}{f_s}$$

8. The centre of gravity of a sequence is defined by $c = \frac{\sum_{n=-\infty}^{\infty} n\,x[n]}{\sum_{n=-\infty}^{\infty} x[n]}$ and can be used as a measure of the *time delay* of a sequence: Using the properties of the DTFT, find

   (a) an expression for *c* in terms of the DTFT of x[n].

   (b) the value of c for the sequence x[n] that has a DTFT is shown in Figure B.

9. For the sequence $x[n]$ shown in Figure C, evaluate the following integral

$$\int_{-\pi}^{\pi} \left|\frac{d}{d\theta}X(\theta)\right|^2 d\theta.$$

